Project Management Plan

Aerospace Transport

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Project Management Plan

Introduction (Dezso):

The aerospace launchpad project requires a scope management plan to define its technical boundaries and project limitations. Launch pad construction involves multiple complex elements, and stakeholder expectations vary widely. The plan outlines essential change management procedures and basic procedures to meet space industry capacity standards while fulfilling all regulatory requirements and operational standards.

Project Scope Statement (Dezso):

Our aerospace project aims to build highly advanced and durable launchpads to cut on overhead costs, addressing the urgent capacity constraints that currently slow down space missions nationwide for commercial and government agencies. Current launch facilities are facing delays exceeding one year due to high demand, which causes delays of more than a year. This plan defines the project scope, outlining important technical specifications. Clear measures for success ensure transparency and alignment among our stakeholders. This project includes multiple durable launchpads, a lot of infrastructure support, and efficient, durable systems designed to speed up turnaround times and boost launch frequency while upholding strict safety standards. Success will be measured and evaluated based on cost-effectiveness, strict adherence to regulatory standards, and execution of multiple demo launches, demonstrating the facility's preparedness and efficiency.

Problem & Need Statement (Christian and Ryan):

The launchpad shortage is really becoming a headache for the entire space industry. Those 12-18 month delays aren't just numbers on a spreadsheet—they're stalling scientific breakthroughs, holding back businesses, and even impacting national security projects. It's pretty frustrating that our existing infrastructure just can't keep up. Florida and Texas sites are completely maxed out, and many were built with only the big rockets in mind, not the smaller vehicles that are becoming so popular. Then there's the weather problem—we're still mainly launching from the same places we've always used, which means we're at the mercy of the same weather patterns and limited to the same flight paths. If we don't tackle this soon, we're essentially putting a speed limit on the entire space industry. The demand is clearly there, but the bottleneck of available launchpads is like trying to funnel highway traffic through a single toll booth.

Project Goal(s) & Objective(s) (Dezso):

We aim to build more launchpads, solve the current shortage affecting commercial and government sectors, and construct new launchpads in specific locations with better operational capabilities. We also aim to reduce delays and infrastructure limitations currently restricting space exploration. These initiatives will help expand the global space industry, allowing more frequent satellite launches, scientific research opportunities, and enhanced national security. The launch pads are designed for a more flexible and practical launch, ensuring they stay valuable and adaptable as technology changes.

Objectives:

- ➔ Double the number of rocket launches possible within the first year to reduce current delays.
- ➔ Reduce launch preparations by **30%** by simplifying processes and designing more practical facilities.
- ➔ Reach **95%** reliability in launch schedules, packing better locations using more accurate weather tracking tools to reduce cancellations.
- ➔ Use renewable energy for **40%** of facility operations to be more environmentally friendly.
- ➔ Maximize equipment reliability by **99.5%**, minimizing the need for maintenance.
- ➔ Design more flexible facilities to handle crew and uncrewed missions, making them useful for many missions.
- ➔ A logistical infrastructure, including rocket fuel storage, parts storage, and transportation networks, to improve the supply chain by **40%**

Deliverable(s) and Feature (Dezso):

Our Aerospace launch pad project will produce the following core deliverables and the associated features:

- ➔ Launch Pad Infrastructure
  - ◆ Reinforced concrete to handle rockets weighing up to 2,000 tons (4,000 KIPS).
  - ◆ Flexible mounting systems that work with different rocket sizes.
  - ◆ Reinforced flame trenches equipped with water-cooling systems to redirect heat.
  - ◆ Lighting protection towers and weather monitoring systems.

- Logistics Infrastructure
    - The fuel storage and distribution setup includes cryogenic, hypergolic, and solid fuel types.
    - Facilities for spare components.
    - Reinforced roads are explicitly designed to handle heavy transportation.
    - Loading and unloading equipment for oversized rocket components.
    - Inventory tracking system that automatically updates stock and reorders supplies.
- Operational Systems
    - Radar and safety-tracking systems for launches.
    - Software to coordinate launch schedules.
    - Track environmental conditions to ensure compliance with FAA regulations.
    - Sensors and automated detection networks with AI-enchanted analysis.
    - Weather prediction for optimal launch windows.
    - Data collection to improve operations over time.
- Site Development
    - Roads connecting directly to state highways.
    - Backup power, water, and communication systems.
    - Secured site perimeter with security checks & monitoring.
    - Drainage and environmental protection measures to minimize ecological impacts.

Success Criteria (Dezso)

- Schedule Compliance
  - Finish construction and have all launch pads ready by August 15, 2025.
- Budget Performance
  - Keep costs within **10%** of the **$350** million budget, documenting and approving any changes.
- Regulatory Compliance
  - Obtain all permits and approval from the FAA, EPA, and OSHA.
- Operational Capability Demonstration
  - Launch at least three rockets (small, medium, heavy) within 60 days.
- Safety Standards
  - Complete the project without any significant safety incidents.
- Environmental Impact
  - Stay within the expected environmental impact levels outlined at the project's start.
- Quality Assurance
  - Inspection with zero critical defects, ensuring all launch systems perform at or above required standards.
- Logistics Performance
  - Logistics systems efficiently support fuel supply, transportation of components, and inventory control.

Project Scope Key Takeaways (Dezso)

1. The project aims to reduce launch delays by doubling launch pad capacity
2. Improved facility designs will reduce preparation time, making launches more frequent and efficient with fewer cancellations
3. Scheduling and sustainability priorities are achieved by careful site selection and renewable energy.

Project Exclusions

Introduction (Dezso)

Project exclusions clearly define what will not be part of the project, protecting the team from unplanned work and helping align stakeholder expectations. Without explicit exclusions, projects risk scope creep and misalignment with planned goals.

Project Exclusions for Aerospace Transport Project (Dezso)

The aerospace transport launchpad building process will focus intensively on ground infrastructure. We have deliberately drawn lines around the several areas maintained to the project's integrity. The rocket and the spacecraft development will remain outside our view, with rocket manufacturing falling into the company's hands. Though our team will handle the initial logistics, our staff won't be involved with operational positions in this phase. The responsibilities will be transferred to the facility operator after completing the project. Any maintenance that arises after the 12-month warranty period will require separate arrangements. Commercial elements such as the visitor center or tourism attraction will not be under construction or will not be necessary for the project. The road development extends to only essential facilities rather than transportation networks. Regarding specialized facilities for sensitive payloads, the deep space communication systems will fall beyond our development scope.

Project Priorities

Introduction (Christian)

Our goal for this project is to set priorities to ensure that when things get tough, we know what matters most. Facing the truth that time, scope, and cost are almost never going to stay perfectly balanced throughout this whole project. We need to have goals set in place in case things go wrong or turn upside down. When we're clear upfront about what's most important, it gives everyone a compass for making decisions when we're under pressure and resources are stretched thin. Too many projects go without having this conversation early, and it can be the main reason for large delays and project failures. Without these guardrails, you end up with different stakeholders pulling in various directions. The engineering team thinks quality is non-negotiable, the finance department is focused on the budget, and the client is fixated on the deadline. Then decisions get made that contradict each other, and the whole project suffers. It's frustrating for everyone and ultimately undermines what we're trying to accomplish. So let's have the hard conversation now about what gives first when push comes to shove. It'll save us a ton of headaches down the road.

Project Priorities for Aerospace Transport (Dezso)

Developing our launchpads. Aerospace operations require strict safety protocols because failures result in catastrophic outcomes that violate technical requirements and safety standards. The industry's critical capacity shortages force us to maintain a tight timeline, which means rushing specific tasks could delay our project. Our project maintains discipline, but cost considerations rank third in our hierarchy, while we make reasonable financial adjustments to preserve quality standards and schedule commitments.

Priority Matrix (Dezso)

Constraint	**Priority Ranking**
Time	1 (Highest)
Scope	2 (Medium)
Cost	3 (Lowest)

Key Takeaways Regarding Project Priorities (Dezso)

The safety and functionality are the non-negotiable elements in our launch pad construction, with the commitment to our time being above all other considerations. Our project delivery time is acknowledged, as following behind can have massive effects. The facilities will remain incomplete monthly, meaning dozens of critical missions will be grounded.

Scope Breakdown

Work Breakdown Structure (WBS) Introduction (Dezso)

The work breakdown structure, known as the WBS, is a backbone for our Aerospace Transport project. Unlike traditional task lists, the WBS decomposes complex work into manageable components. The WBS organizes project tasks into more minor elements until they reach work packages that can be effectively estimated and assigned. This method ensures nothing falls through the cracks while creating natural accountability points throughout the project's lifecycle.

Our WBS helps turn the launch pad development from an overwhelming project into a series of achievable tasks. This helps give the stakeholders a more precise visualization of the project scope.

WBS Introduction (Dezso)

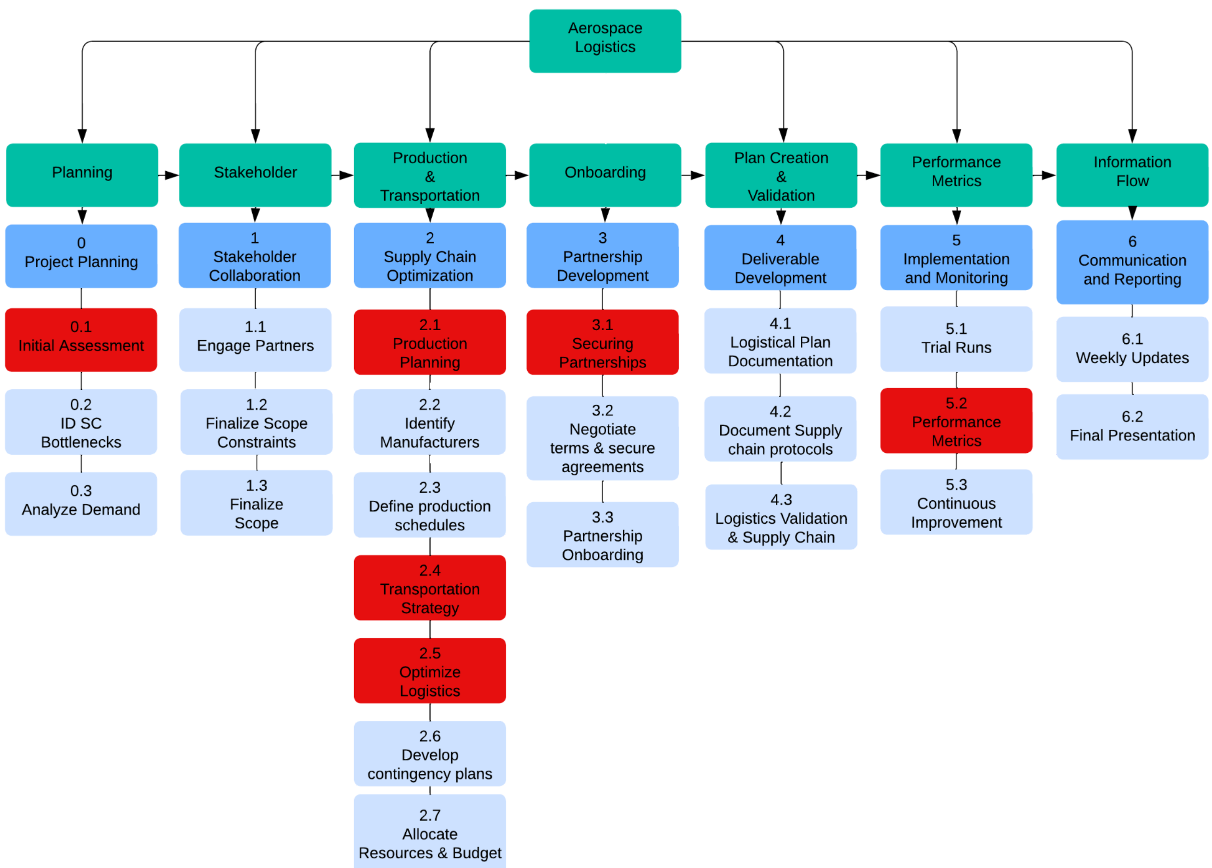
The Aerospace transport WBS consists of seven major summary tasks divided into 24 work packages. The structure follows the project lifecycle from foundational planning activities to communication and reporting.

Each of the summary tasks contains between three and seven work packages. Production and transportation have the most complex packages, with seven work packages each. We have used a numbering indexing system. Starting on Day 0—Day 6, major summary tasks will be done with decimal points indicating subordinating work packages to create a precise reference point for all the project elements.

The critical path is highlighted in red, including these task elements, which demand special attention as they directly impact the project timeline and success.

WBS Chart (Dezso)

The WBS chart displays our project's flowchart structure. The primary summary tasks appear in teal at the top level, with a numerical designation and descriptive titles directly below in blue. The critical path items are highlighted in red to emphasize their importance to the project's success.



WBS Dictionary Introduction (Dezso)

The WBS chart we made provides a structured visualization of the project, while the WBS Dictionary is a very important document displaying detailed descriptions for each component. It turns the high-level framework into guidance by defining specific deliverables, scope boundaries, acceptance criteria, and key expectations for each work package. For our project team members, the WBS Dictionary functions as the source for scope definition, ensuring clarity not only on the work being performed but also on the criteria for each work package's completion. A comprehensive dictionary developed to define the work packages is shown below.

WBS Dictionary (Dezso)

Task ID	Work Package	Definition
0.1	Initial Assessment	Assess the logistics, identify the gaps, and establish baseline metrics for improvement. The deliverables include a comprehensive assessment report and a presentation.
0.2	ID SC Bottlenecks	Analyze supply chain constraints affecting launch pad development, including materials sourcing, specialized equipment, and transportation.
0.3	Analyze Demand	Industry trends, stakeholder input, and historical data will be analyzed to predict future launch pad capacity requirements. The demand projection will include a five-year growth report.
1.1	Engage Partners	Build a relationship with the aerospace industry, including manufacturers and regulatory bodies. Create a signed engagement and partnership documentation.

Task ID	Work Package	Definition
1.2	Finalize Scope Constraints	The project boundaries should be defined through collaborative stakeholder workshops, the inclusions and exclusions should be formalized, and a scope agreement document should be delivered with stakeholder acknowledgment.
1.3	Finalize Scope	The project scope constraints transform into a complete scope statement, including specific deliverable specifications. The process results in approved final scope documentation for all stakeholders.
2.1	Production Planning	Create a manufacturing and procurement schedule plan for specialization in launch pad components, including concrete, steel reinforcements, and control systems. Delivering integrated production schedules with critical milestones.
2.2	Identify Manufactures	The goal is to identify and choose multiple suppliers of specialized aerospace-grade materials and components through a thorough valuation process. It generates an approved vendor list with all the capability assessments.
2.3	Define Production Schedules	Establish production schedules for manufacturing, testing, and delivery that match the project timeline. This will provide an integrated production timeline with mapped dependencies.
2.4	Transportation Strategy	Create a dedicated plan with procedures for oversized and sensitive aerospace components, including route planning, equipment requirements, and handling

Task ID	Work Package	Definition
		protocols. Develop a complete transportation plan.
2.5	Optimize Logistics	To reduce delays, damage risk, and costs, the transportation and handling processes should be refined through simulations and industry best practices. It delivers optimization reports with implementation recommendations.
2.6	Develop Contingency Plans	Develop response strategies for potential supply chain disruptions, including alternative sourcing, route diversification, and recovery options—risk response documentation with triggering criteria.
2.7	Allocate Resource & Budget	Assign financial teams across all supply chain activities based on criticality and complexity. Delivering resource allocation with justifications.
3.1	Securing Partnership	The company establishes relationships with aerospace industry stakeholders through negotiated agreements that outline responsibilities, contribution terms, and benefit distribution. The company delivers signed partnership agreements.
3.2	Negotiate Terms & Secure Agreements	Create comprehensive contracts with all partners, including performance metrics, incentives, and dispute resolution mechanisms. Distribute executed contracts to all key suppliers.
3.3	Securing Partnership	The company establishes relationships with key aerospace industry stakeholders by negotiating agreements outlining responsibilities, benefits, and contributions, as well as delivering signed contracts.

Task ID	Work Package	Definition
4.1	Logistics Plan Documentation	Develop a detailed documentation system that covers all logistical aspects of launch pad development requirements and standards, delivering logistical documentation.
4.2	Document Supply Chain Protocols	Processing information for all supply chain activities, including quality control, transportation requirements, and material handling. Documentation is delivered with workflow diagrams.
4.3	Logistics Validation & Supply Chain	The process includes testing and validating systems and processes through simulation testing, limited-scale implementation, and the delivery of validation reports with performance metrics.
5.1	Trial Runs	Limited testing of essential logistical components must be performed to confirm their operation and discover potential enhancements. The production process includes creating a trial performance analysis together with recommendation logs.
5.2	Performance Metrics	Create a measurement system for all logistical operations to track time, cost, quality, and reliability indicators for accuracy and deliver a metric dashboard with baseline performance data.
5.3	Continuous Improvement	The implementation follows an ongoing improvement process that uses performance data analysis and stakeholder feedback. The system generates an improvement roadmap containing prioritized initiatives.

Task ID	Work Package	Definition
6.1	Weekly Updates	The company will provide regular status reports to all stakeholders about logistics operations, challenges, and progress. Standardized weekly reporting templates and distribution systems will also be delivered.
6.2	Final Presentation	The report includes a summary of completed logistical implementation, performance achievements, and recommendations for future operations. It also contains a detailed closeout report and presentation.

WBS Dictionary Key Takeaways (Dezso)

The detailed dictionary provides an understanding of our project's logistics methods. The validation and testing process appears throughout various work packages. This approach makes the essential role of reliability in aerospace operations evident because a logistics failure would endanger missions that cost millions of dollars. The progressive development from planning through implementation demonstrates our methodological approach to aerospace logistics, which builds each component from previous work to maintain system integrity.

The detailed mapping establishes a disciplined execution framework that addresses all critical elements because minor oversights in this project would result in major consequences for launch capabilities and operations.

Schedule Management Plan

Project Schedule Introduction (Ryan)

The purpose of creating and managing a project schedule is to show how the team will manage to plan the schedule and be able to manage it. For such a complicated task like helping construct aerospace launchpads, there must be an accurate schedule so everything can be done quickly and reasonably. This will help us align our goals with stakeholders' and ensure no cost delays. This also plays a massive part in helping us find the critical path, affecting our project's completion deadline.

Tasks & Activities Introduction (Ryan)

Before a solid schedule is built, we must divide the work into specific tasks and determine how long each will take. This will help us figure out how long each one will take by estimating the activity's duration, and it will be a key part of the planning process. Each task will relate to the WBS, align with our schedule, and help others identify our predecessors. This means whether we finish on time or not. To ensure we came up with realistic time estimates, we used a mix of many different things, like experts' opinions, past projects like this one, and a lot of industry averages.

Tasks & Activities Chart (Dezso)

ID	Work Package	Duration	Predecessor
0.1	Initial Assessment	12	
0.2	Identify SC Bottlenecks	2	
0.3	Analyze Demand	6	0.2
1.1	Engage Partners	4	0.3

ID	Work Package	Duration	Predecessor
1.2	Finalize Scope Constraints	2	1.1
1.3	Finalize Scope	3	1.2
2.1	Production Planning	12	0.1
2.2	Identify Manufactures	3	1.3
2.3	Define Production Schedules	6	2.2
2.4	Transportation Strategy	12	2.1
2.5	Optimize Logistics	12	2.4
2.6	Develop Contingency Plans	4	2.3
2.7	Allocate Resource & Budget	1	2.6
3.1	Securing Partnerships	12	2.5
3.2	Negotiate Terms & Secure Partnerships	5	3.1
3.3	Partnerships Onboarding	2	3.2
4.1	Logistical Plan Documentation	2	3.3
4.2	Document Validation & Supply Chain	2	4.1
4.3	Logistics Validation & Supply Chain	3	4.2
5.1	Trail Runs	6	4.3
5.2	Performance Metrics	12	3.1

ID	Work Package	Duration	Predecessor
5.3	Continuous Improvement	5	5.2
6.1	Final Presentation	1	5.3

Key Takeaways for Duration Estimates (Christian)

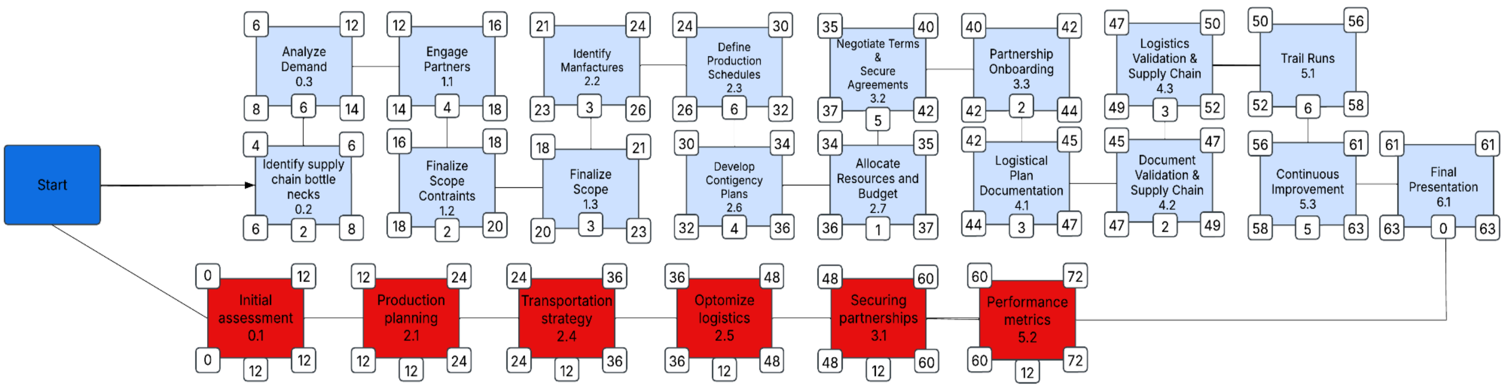
We based our task durations on expert opinion, industry averages, and past data from similar projects of this nature. This helped us obviously make sure it was an achievable and realistic goal. We also carefully sequenced the activities with a clear view of project efficiency, and it will help us stay aligned with our deadline.

Project Network Diagram & Critical Path

Network Diagram Introduction (Christian)

The Network Diagram is one of the most valuable tools in this project. It helps visualize the sequence and dependencies of activities within our aerospace launchpad project. It represents the project's workflow, highlighting the relationships between tasks and identifying the critical path. Using the Network Diagram, we can effectively plan, coordinate, and monitor the project's progress, ensuring that all activities are completed in a timely and efficient manner.

Network Diagram (Dezso)



Network Diagram Chart (Ryan)

Task	ES/EF LS/LF	Predecessor
0 Project Start	**0 Days** Early Start/Finish - **0/0** Late Start/Finish - **0/0**	Project Start
0.1 Initial Assessment	**12 Days** Early Start/Finish - **0/12** Late Start/Finish - **0/12**	Delay - 0 Predecessor - 0 Type - N/A
0.2 Identify SC Bottlenecks	**2 Days** Early Start/Finish - **4/6** Late Start/Finish - **6/8**	Delay - 0 Predecessor - 0 Type - N/A
0.3 Analyze Demand	**6 Days** Early Start/Finish - **6/12** Late Start/Finish - **8/14**	Delay - 6 Predecessor - 0.3 Type - Finish-Start
1.1 Engage Partners	**4 Days** Early Start/Finish - **12/16** Late Start/Finish - **14/18**	Delay - 4 Predecessor - 0.2 Type - Finish-Start

Task	ES/EF LS/LF	Predecessor
1.2 Finalize Scope Constraints	**2 Days** Early Start/Finish - **16/18** Late Start/Finish - **20/23**	Delay - 2 Predecessor - 1.1 Type - Finish-Start
1.3 Finalize Scope	**3 Days** Early Start/Finish - **18/21** Late Start/Finish - **20/23**	Delay - 2 Predecessor - 1.2 Type - Finish-Start
2.1 Production Planning	**12 Days** Early Start/Finish - **12/24** Late Start/Finish - **12/24**	Delay - 12 Predecessor - 0.1 Type - Finish-Start
2.2 Identify Manufacturers	**3 Days** Early Start/Finish - **21/24** Late Start/Finish - **23/26**	Delay - 3 Predecessor - 1.3 Type - Finish-Start
2.3 Define Production Schedule	**6 Days** Early Start/Finish - **24/30** Late Start/Finish - 26/32	Delay - 3 Predecessor - 2.2 Type - Finish-Start
2.4 Transportation Strategy	**12 Days** Early Start/Finish - **24/36** Late Start/Finish - **24/36**	Delay - 12 Predecessor - 2.1 Type - Finish-Start
2.5 Optimize Logistics	**12 Days** Early Start/Finish - **36/48** Late Start/Finish - **36/48**	Delay - 12 Predecessor - 2.4 Type - Finish-Start
2.6 Develop Contingency Plans	**4 Days** Early Start/Finish - **30/34** Late Start/Finish - **32/36**	Delay - 6 Predecessor - 2.3 Type - Finish-Start
2.7 Allocate Resources & Budget	**1 Days** Early Start/Finish - **34/35** Late Start/Finish - **36/37**	Delay - 4 Predecessor - 2.6 Type - Finish-Start
3.1 Securing Partnerships	**12 Days** Early Start/Finish - **48/60** Late Start/Finish - **48/60**	Delay - 12 Predecessor - 2.5 Type - Finish-Start
3.2 Negotiate Terms & Secure Agreements	**5 Days** Early Start/Finish - **35/40** Late Start/Finish - **37/42**	Delay - 12 Predecessor - 3.1 Type - Finish-Start

Task	ES/EF LS/LF	Predecessor
3.3 Partnerships Onboarding	**2 Days** Early Start/Finish - **40/42** Late Start/Finish - **42/44**	Delay - 5 Predecessor - 3.2 Type - Finish-Start
4.1 Logistical Plan Documentation	**3 Days** Early Start/Finish - **42/45** Late Start/Finish - **47/49**	Delay - 2 Predecessor - 3.3 Type - Finish-Start
4.2 Document Validation & Supply Chain	**2 Days** Early Start/Finish - **45/47** Late Start/Finish - **47/49**	Delay - 3 Predecessor - 4.1 Type - Finish-Start
4.3 Logistics Validation & Supply Chain	**3 Days** Early Start/Finish - **47/50** Late Start/Finish - **49/52**	Delay - 2 Predecessor - 4.2 Type - Finish-Start
5.1 Trial Runs	**6 Days** Early Start/Finish - **50/56** Late Start/Finish - **52/58**	Delay - 3 Predecessor - 4.3 Type - Finish-Start
5.2 Performance Metrics	**12 Days** Early Start/Finish - **60/72** Late Start/Finish - 6**0/72**	Delay - 12 Predecessor - 3.1 Type - Finish-Start
5.3 Continuous Improvement	**5 Days** Early Start/Finish - **56/61** Late Start/Finish - **58/63**	Delay - 12 Predecessor - 5.2 Type - Finish-Start
6.1 Weekly Updates	**N/A Days** Early Start/Finish - **0/61** Late Start/Finish - **0/63**	N/A
6.2 Final Presentation	**0 Days** Early Start/Finish - **61/61** Late Start/Finish - **63/63**	Delay - 5 Predecessor - 5.3 Type - Finish-Start

Network Diagram Key Takeaways (Ryan)

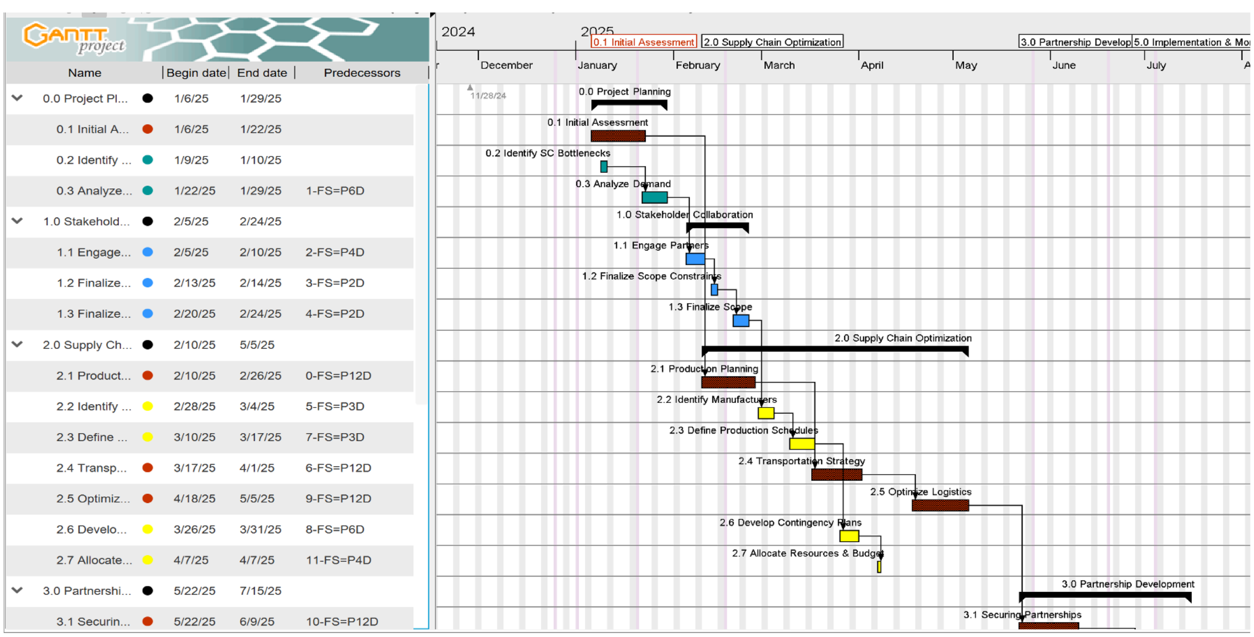
Identifying our critical path allowed us to see which activities will finish on time to avoid a delay in our project. This will help us focus on what matters most. This visual also helped us visualize some bottlenecks we may have and helped us have a smooth project.

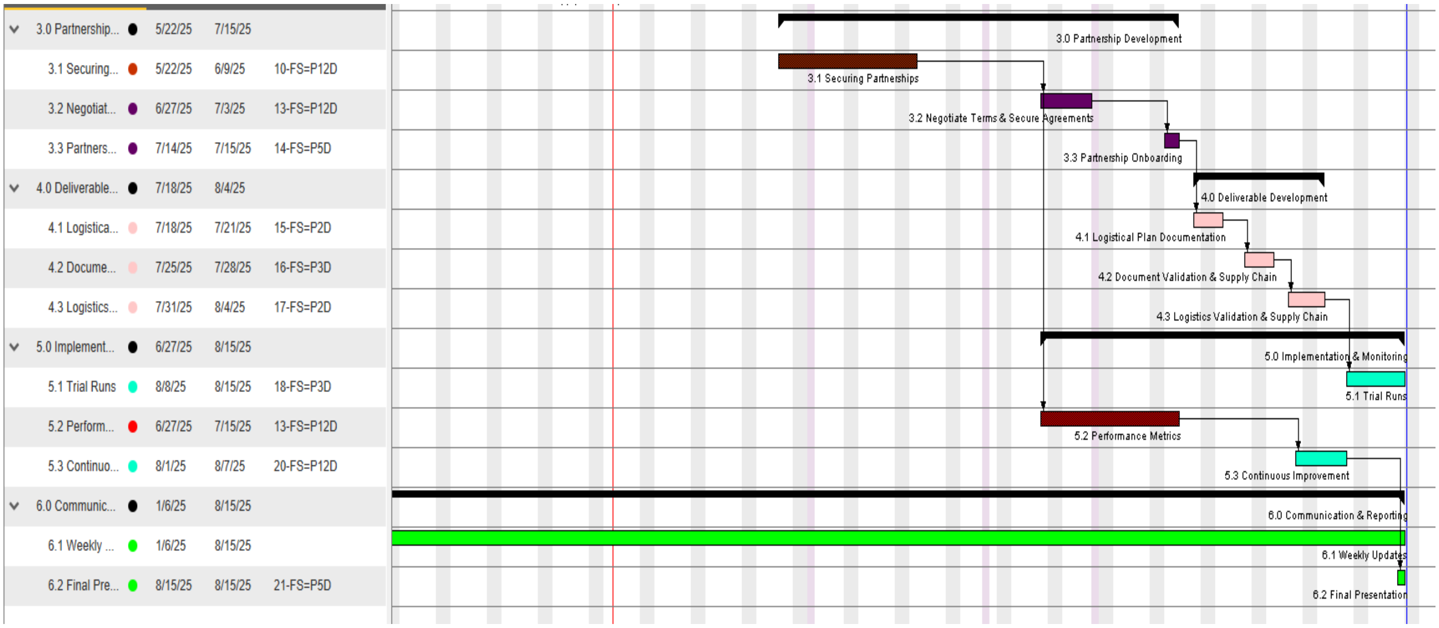
Project Schedule

Project Schedule Introduction (Christian)

One of the most critical components of our Project plan is the Schedule/Gantt Chart, which describes the timeline and order of tasks needed to accomplish our goals. A clear plan and schedule are necessary for a complicated project like building aerospace launchpads to guarantee that all functions are finished correctly and on schedule. The schedule helps us juggle our resources, too. We only have so many people, so much equipment, and definitely a limited budget. Having everything mapped out lets us make sure we're using what we have effectively. When we stick to the plan, we avoid those frustrating delays that always seem to cascade into bigger problems. I've been on projects without good scheduling before, and honestly, it was a nightmare..

Project Schedule Chart (Ujas)





Project Schedule Key Takeaways (Dezso)

The 171-day project showed that stakeholder activities during early phases establish essential buffers that protect technical work in later stages. For logistics optimization, the project needed more labor hours than planned, and with manufacturing delays and weather disruptions, documentation tasks had to operate on compressed schedules. The critical path activities 2.4 and 2.5 proved most sensitive to schedule changes because they needed weekend work to stay on schedule.

The project's August 15th deadline involved the strategic deployment of contingency resources during final testing, which is needed to maintain schedule reserves.

Cost Management Plan

Project Budget Intro (Christian)

The Project Budget outlines the financial resources needed for our aerospace launchpad construction. It ensures all aspects of the project are funded appropriately, helping us avoid cost overruns and maintain financial stability. By adhering to the budget, we can make informed decisions and keep the project on track.

Cost Estimating

Sources & Techniques (Dezso)

For our project, we have developed a budget by combining historical data, industry benchmarks, expert input, and looking at published indexes—the estimation techniques included from the past project, bottom-up estimating for the critical path work packages, three-point estimating for the uncertain areas, and vendor quotes analysis. This mix of the source techniques helps create the most realistic cost baseline.

Cost Estimate Chart (Christian)

Work Package	Labor Cost	Material Cost	Equipment Cost	Total Cost
0.1 Initial Assessment	18200	3500	1800	23500
0.2 Identify SC Bottlenecks	7000	1200	0	8200
0.3 Analyze Demand	7280	4500	0	11780
1.1 Engage Partners	12000	2500	1000	15500
1.2 Finalize Scope Constraints	8500	1800	500	10800

Work Package	Labor Cost	Material Cost	Equipment Cost	Total Cost
1.3 Finalize Scope	9000	2000	700	11700
2.1 Production Planning	10000	3000	1200	14200
2.2 Identify Manufacturers	7500	1500	600	9600
2.3 Define Production Schedule	8000	2200	800	11000
2.4 Transportation Strategy	6500	1800	500	8800
2.5 Optimize Logistics	7000	2000	700	9700
2.6 Develop Contingency Plans	5500	1500	400	7400
2.7 Allocate Resources & Budget	9500	2500	1000	13000
3.1 Securing Partnerships	8000	2000	600	10600
3.2 Negotiate Terms & Secure Agreements	7500	1800	500	9800
3.3 Partnerships Onboarding	6000	1500	400	7900
4.1 Logistical Plan Documentation	5500	1200	300	7000
4.2 Document Validation & Supply Chain	6500	1500	400	8400
4.3 Logistics Validation & Supply Chain	7000	1800	500	9300
5.1 Trial Runs	8000	2000	600	10600
5.2 Performance Metrics	6500	1500	400	8400
5.3 Continuous Improvement	7500	1800	500	9800

Work Package	Labor Cost	Material Cost	Equipment Cost	Total Cost
6.1 Weekly Updates	5000	1200	300	6500
6.2 Final Presentation	6000	1500	400	7900

Cost Estimate Key Takeaways (Christian)

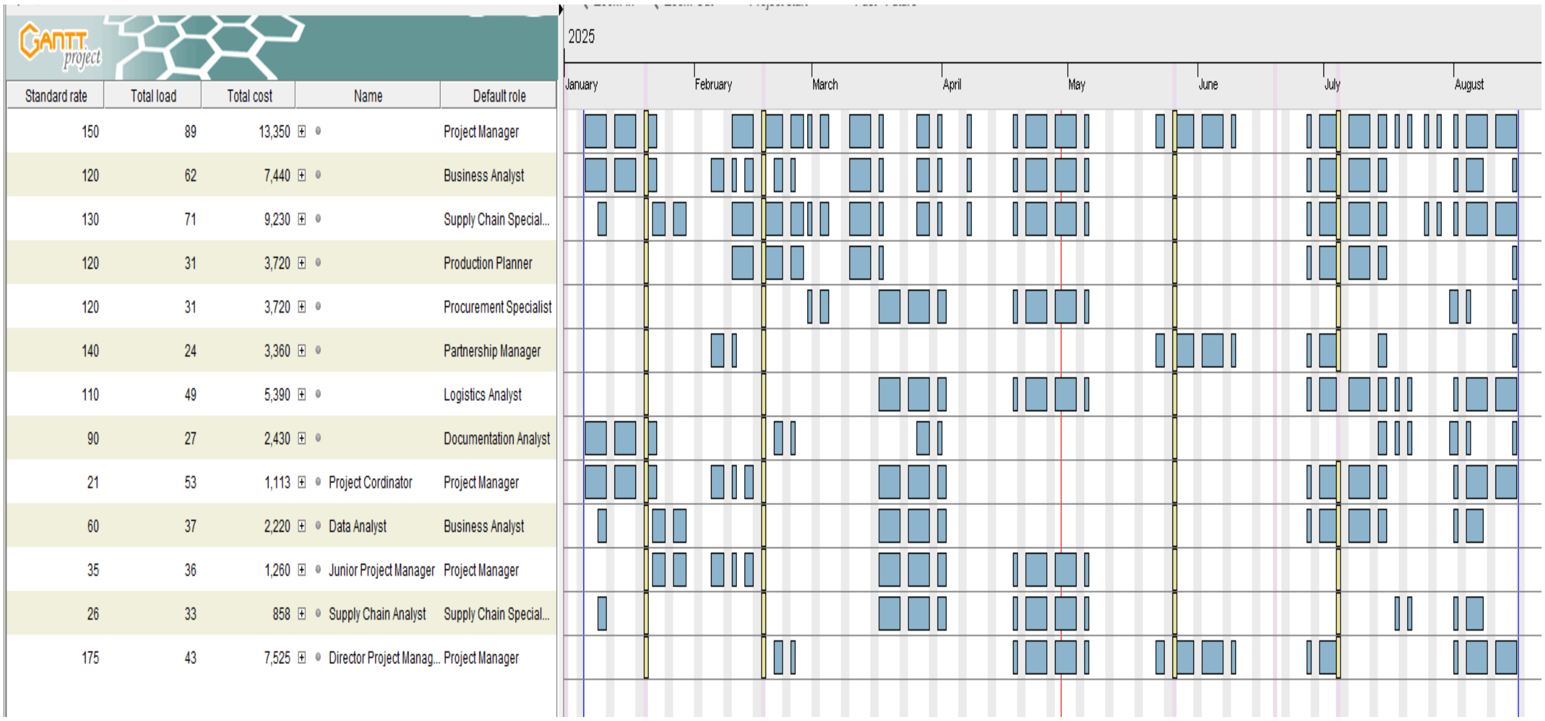
By using a mix of experts in this field's advice, past data from projects similar to this one, and a three-point estimating system. We used all of these to build the budget chart that is accurate and not going to make us feel surprised or shocked about the cost later on in the project. We also put high value on the critical path and made sure those parts got the most funding.

Human Resource Management Plan

Sources & Techniques (Christian)

Our approach to human resource management requires strategic optimization of team members and their distribution across the entire project lifecycle. The work breakdown structure analysis determines resource needs, while allocation decisions depend on the expertise requirements of essential work packages. The team members who track project budget through the schedule create a visualization that displays their work distribution from January through August 2025. The detailed labor budget shows a total cost of $61,616.04 distributed across 171 working days with 3,610 labor hours allocated to essential experts at critical points to maintain project efficiency.

Human Resource Chart (Fabian)



Human Resource Allocation (Dezso)

This approach was used to assign the resources or "roles" to the work packages, balancing the need for specialized expertise with the demands within the critical path. Full-time employees are transferred to the essential activities of the path (Initial Assessment, Production Planning, Transportation strategy), and can maintain momentum to minimize delays.

The roles requiring more in-depth knowledge, such as supply chain specialist and logistical analyst, are focused on contributions closely related to work packages, avoiding any issues across unrelated tasks. Able to maximize the resource efficiency, including cross-functional roles for the Project Manager, allowing them to serve as task owners and integrators across multiple working streams. The specialized team is deployed using a time-phase approach, enabling them to focus on their efforts during the highest demand periods without requiring full-time commitment to the project. We have also added junior and senior team members, with senior staff concentrating on critical decision points while mentoring junior colleagues, to prevent employee burnout and maintain work quality. Workload planning ensures that no team member exceeds 40 allocated hours per week. When resource allocations had challenges emerged, we addressed them by adding more people on the team for the non critical tasks to balance workloads, temporarily increasing the business analyst capacity during february intensive planning phase, assigning logistical analyst to sequential rather than concurrent functions during periods of high demand which established points between a supply chain specialist and procurement specialist to ensure workflow.

Human Resource Key Takeaways (Dezso)

Our resource allocation strategy depends on project priorities, which directly affect critical success factors. The budget allocates more than $53,000 for specialized expertise in transportation logistics optimization and performance measurement through essential path activities.

The project maintains balanced resource distribution through sequencing and resource planning while keeping resource loading consistent. This approach minimizes the chance of overallocation because it helps prevent increased costs and reduced quality. The cross-functional team plan, together with effective knowledge transfer, enables risk identification and improves communication between project teams.

Also, the resources for the work packages with significant matters cost 2.4 and 2.5, which ensures that major investments are being overseen. The project budget of $137,697 has been optimized by using high-cost resources at times of maximum value and cost-effective resources for work activities. This approach helps to keep a level that supports the successful delivery of the project.

Time-Phased Budget

Time-Phased Budget Introduction (Ujas)

The time-phased budget for this project provides a detailed projection of costs and labor hour allocations over the full project lifecycle, aligned with each scheduled activity. Activities such as Initial Assessment, Production Planning, Transportation Strategy, Optimize Logistics, and Performance Metrics are scheduled between January and August 2025, with start dates strategically phased to minimize resource overlap and ensure efficient workflow. The budget also shows a strong link between planned labor utilization and activity deliverables, helping ensure financial control and performance monitoring across all project phases. This phased approach allows the project management team to proactively manage cash flow, anticipate funding needs, and track actual performance against planned targets.

Time-Phased Budget Table (Dezso & Ujas)

ID	Activity	Predecess	Duration in day	Reference 1	Reference 2	Reference 3	Start Dat	Januar	Feburary	March	April	May
0.1	Initial Assessment	N/A	23	Project consultants	Risk analysis	Industry baseline supports	1/6/2025	$4,572.00	$ -	$ -	$ -	$ -
0.2	Identify SC Bottlenecks	N/A	1	Bottleneck Auditors	Operational Flow Experts	Process Mapping Tools	1/9/2025	$ 432.00	$ -	$ -	$ -	$ -
0.3	Analyze Demand	0.2	7	Market Analysts	Demand Forecasting Software	Consumer Reports	1/22/2025	$1,350.00	$ -	$ -	$ -	$ -
1.1	Engage Partners	0.3	5	Stakeholder Managers	Outreach Coordinators	Supplier Directories	2/5/2025	$ -	$ 1,264.00	$ -	$ -	$ -
1.2	Finalize Scope Constraints	1.1	1	Project Planners	Legal Advisors	Constraint Analysis Tools	2/5/2025	$ -	$ 352.00	$ -	$ -	$ -
1.3	Finalize Scope	1.2	4	Legal Advisors	Project Planners	Stakeholder Managers	2/13/2025	$ -	$ 1,155.00	$ -	$ -	$ -
2.1	Production Planning	0.1	16	Production engineers	Inventory systems	ERP software engineers	2/20/2025	$ -	$ 4,800.00	$ -	$ -	$ -
2.2	Identify Manfacturers	1.3	5	HR Consultants	Training Coordinators	Job Role Guidelines	2/28/2025	$ -	$ 1,200.00	$ -	$ -	$ -
2.3	Define Production Schedules	2.2	7	Resource Managers	Task Scheduling Software	Workload Balancers	3/10/2025	$ -	$ -	$ 3,120.00	$ -	$ -
2.4	Transportation Strategy	2.1	15	Supply chain analyst	Carrier Networks	Route optimization tools	3/17/2025	$ -	$ -	$ 4,464.00	$ -	$ -
2.5	Optimize Logistics	2.4	17	Logistics consultants	Simulation Software	Distribution Network Reports	4/18/2025	$ -	$ -	$ -	$ 10,392.00	$ -
2.6	Develop Contingency Plans	2.3	5	Policy Consultants	Legal Advisors	Consumer Reports	3/26/2025	$ -	$ -	$ 1,960.00	$ -	$ -
2.7	Allocate Resource & Budget	2.6	0	Legal Advisors	Stakeholder Managers	Vendor Management Team	4/7/2025	$ -	$ -	$ -	$ 400.00	$ -
3.1	Securing Partnerships	2.5	18	Vendor Management Team	Legal Advisors	Supplier Databases	5/22/2025	$ -	$ -	$ -	$ -	$5,580.00
3.2	Negotiate Terms & Secure Agreements	3.1	6	Policy Consultants	Compliance Auditors	Quality Assurance Manuals	6/27/2025	$ -	$ -	$ -	$ -	$ -
3.3	Partnership Onboarding	3.2	1	Reporting Analysts	Dashboard Tools	Executive Reporting Templates	7/14/2025	$ -	$ -	$ -	$ -	$ -
4.1	Logisitical Plan Documentation	3.3	3	Tech Support	Data Migration Tools	System Integrators	7/18/2025	$ -	$ -	$ -	$ -	$ -
4.2	Document Validation & Supply Chain	4.1	3	IT Auditors	Security Analysts	Firewall Vendors	7/25/2025	$ -	$ -	$ -	$ -	$ -
4.3	Logistics Validation & Supply Chain	4.2	3	Backup Providers	Cloud Storage Vendors	Redundancy Planners	7/31/2025	$ -	$ -	$ -	$ -	$ -
5.1	Trail Runs	4.3	7	Feedback Analysts	Survey Tools	Focus Group Providers	8/8/2025	$ -	$ -	$ -	$ -	$ -
5.2	Performance Metrics	3.1	18	Data Analysts	KPI Dashboards	Business Intelligence Tools	6/27/2025	$ -	$ -	$ -	$ -	$ -
5.3	Continous Improvement	5.2	6	Project Planners	Supply chain analyst	Data Analysts	8/1/2025	$ -	$ -	$ -	$ -	$ -
6.2	Final Presentation	5.3	0	Vendor Management Team	Project Planners	KPI Dashboards	8/15/2025	$ -	$ -	$ -	$ -	$ -
Total			171					$6,354.00	$ 8,771.00	$ 9,544.00	$ 10,792.00	$5,580.00

Key Takeaways (Ujas)

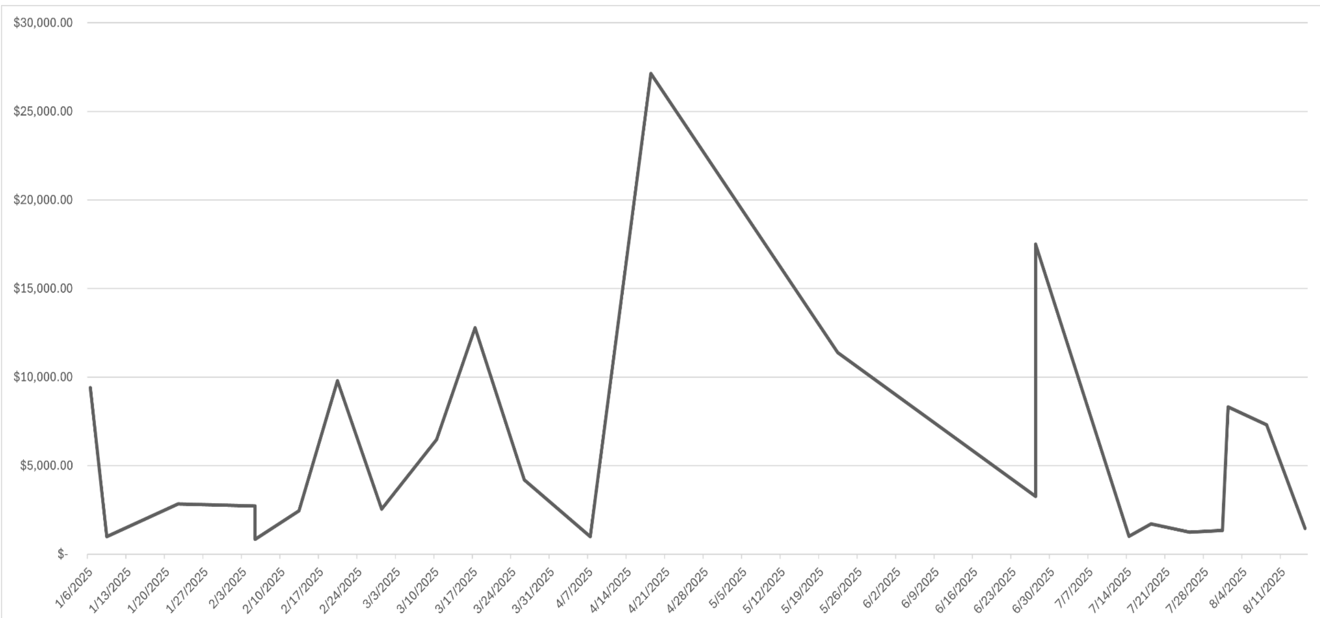
Labor hours total 3,610 hours, and labor costs are estimated at $61,616.04, with material costs adding an additional $9,623.00, bringing the overall projected budget to $137,697.40. Major cost peaks are associated with high-effort activities such as Optimize Logistics in April 2025 ($10,392.00 in labor and $5,824.00 in materials) and Performance Metrics in June 2025 ($8,532.04 in labor). These costs are reflected in the time-phased expenditure curve shown below, where expenditures increase and intensify during periods of heavy project activity.

Budget Line

Budget Line Chart Introduction (Christian)

The Budget Line Chart is a fundamental tool for visualizing the project's financial components. It provides an easy-to-understand view of project expenses, budget distribution, and economic performance throughout different periods. Our project requires detailed and exact budget management to minimize high resource usage and cost overruns. The development of launch pads requires significant financial investment. The budget line chart enables us to distribute and monitor all resources and project expenses.

Budget Line Chart (Dezso)



Key Takeaways Budget Line Chart (Ujas)

The budget line chart shows that project spending begins in January 2025, with small changes through early March as initial project activities are completed. A rise occurs in mid-April, where costs peak near $30,000, reflecting the most resource-intensive phase of the project, likely tied to critical activities such as Optimize Logistics. After this peak, spending declines through May and June as activities slow down. A second, smaller increase appears at the end of June, around $17,000, aligning with additional project critical activities like Performance Metrics. Following this, costs drop increasingly in July and remain low through

August as the project moves toward finalization and closeout. Overall, the project follows a classic ramp-up, peak, and ramp-down cost pattern, with the majority of expenses concentrated in the middle months of the schedule.

Stakeholder Analysis

Stakeholder Analysis Introduction (Christian)

Conducting a stakeholder analysis systematically identifies and understands the interests, expectations, and influence of all parties involved in the project. Ensuring all stakeholders are aligned with the project's goals and objectives is crucial for a complex endeavor like constructing aerospace launchpads. This alignment helps mitigate risks, avoid cost delays, and ensure smooth project execution.

Stakeholder Analysis (Christian)

Stake Holders	Power	Interest	Reasons
Space X	High	Medium	Innovation in space travel, cost-effective launch solutions, expanding market share
Blue origin	High	High	Space exploration, scientific research, technological advancement, international collaboration
Boeing	Medium	High	Aircraft manufacturing, defense contracts, technological innovation
NASA	Low	Low	Sustainable space exploration, commercial space travel, technological development
U.S Government	High	High	Focuses on public-private partnerships, research and development, space priorities framework, climate monitoring, and international partnerships.
Northrop Grumman	Medium	Medium	Defense and aerospace technology, government contracts, innovation in space systems

Project Stakeholders (Christian)

Our project stakeholders include SpaceX, Blue Origin, Boeing, NASA, the U.S. Government, and Northrop Grumman.

Stakeholder Key Takeaways (Ujas)

SpaceX holds high power and medium interest, driven by its focus on innovation in space travel, cost-effective launch solutions, and expanding market share. Blue Origin has both high power and high interest due to its commitment to space exploration, scientific research, and international collaboration. Boeing is identified as having medium power but high interest, given its role in aircraft manufacturing, defense contracts, and technological innovation. NASA shows both low power and low interest, concentrating on sustainable space exploration, commercial space travel, and technological development. The U.S. Government has high power and high interest, reflecting its investment in public-private partnerships, research and development, climate monitoring, and international cooperation. Lastly, Northrop Grumman holds medium power and medium interest, with a focus on defense and aerospace technology, government contracts, and innovation in space systems. These stakeholders influence the project's direction and success through their resources, priorities, and strategic objectives.

Communication Management Plan

Project Communication Needs (Nathan)

The success of project management relies on appropriate communication methods, which are critical for complex and dangerous projects such as aerospace launch pad construction. The project team members, stakeholders, and outside partners will stay synchronized throughout the project duration when communication requirements are identified early and documented. The procedure defines which team members should receive information and when and what format they should receive it. The project achieves better decision-making and reduces redundant work and misunderstandings through the distribution of accurate and current information across all project levels. The project requires effective communication to manage tasks and track progress while controlling expectations in its demanding environment involving engineers, suppliers, regulatory bodies, and senior leadership.

Project Communication (Nathan)

The project communication plan ensures team collaboration through its design while maintaining transparency to keep all members informed and engaged. The project team has engineering, logistics coordinators, and environmental compliance officers participating in weekly status meetings to track progress, resolve impediments, and adjust schedules when needed. To prevent downstream disturbances, the team distributes essential information about technical design changes and equipment acquisition delays through direct communications or meetings. The project team provides executive briefings to high-level stakeholders monthly, including project schedules, strategic risk assessments, and budget use summaries. The project delivers environmental compliance updates to regulatory bodies through quarterly and immediate reports when necessary to fulfill operational and legal requirements.

Communication Plan (Nathan)

Information Type	Sender	Receiver	Frequency/Timing	Method of Communication
Project status updates	Project Manager	Project Team, Stakeholders	Weekly	Email, MS Teams, MS Project

Information Type	Sender	Receiver	Frequency/Timing	Method of Communication
Engineering Design Changes	Technical Lead	Engineering Team, QA, PM	As Needed	Email, Design review meetings
Budget & Financial Reports	Financial Analyst	Project Manager	Monthly	Email, Financial Dashboard
Risk & Issue logging	Risk Manager	PM, Compliance Officer	Bi-weekly or on an urgent basis	Shared document, email
Environmental compliance	Compliance Officer	Government Agencies, PM	Per the regulation schedule	Formal reports, Online portal
Construction progress update	Site Manager	Project manager, engineers	Weekly	On-site reports, team meetings
Launch briefings	Operations Lead	Stakeholders, Project Team	Before major milestones	Presentation, Video Conference
Executive summary reports	Project Manager	Project sponsor, Company Executives	Monthly	Email, Presentation
Stakeholder feedback	External Stakeholders	PMO, Project Sponsor	Quarterly	Surveys, Stakeholder Meetings

Project Communication Key Takeaways (Nathan)

The main lesson learned from creating and implementing this communication plan is that structure and flexibility are essential. A communication strategy needs to be specific enough to handle regular and significant interactions and adaptable to handle unexpected issues or changes in stakeholder requirements. It reinforces accountability, fosters trust among team members, and keeps leadership well-informed, empowering leaders to make faster, more effective decisions. Ultimately, the communication plan is not just a background process—it is a core project management tool that directly influences the project's success by keeping people informed, engaged, and aligned from concept to launch.

Project's Assumptions

Project Assumptions Introduction (Christian & Fabian)

This report outlines the key assumptions for our aerospace launchpad project. We assume timely regulatory approvals, continuous funding, availability of required technologies, and alignment among stakeholders. Additionally, we expect skilled workforce availability, stable supply chains, minimal weather delays, controlled operational costs, growing launch demand, and scalable infrastructure design.

Assumptions (Christian & Fabian)

1. Regulatory Approvals – All necessary FAA and government permits will be secured without significant delays prolonging the overall project completion time. (Stakeholders List & Schedule)
2. Funding Availability – Sufficient funding will be available throughout the project lifecycle. (Stakeholders List & Budget)
3. Technology Readiness – Flexible launch pads and automation technologies are required, available, and feasible. (Project Scope & Resources Chart)
4. Stakeholder Alignment – Government agencies, private partners, and investors will remain aligned on project goals. (Stakeholders List)
5. Workforce Availability – Skilled labor and technical experts will be available for construction and operations. (Resources Chart)

6. Supply Chain Stability – Materials and critical components will be delivered on schedule without significant disruptions. (Schedule & Budget)
7. Weather Conditions – Selected launch sites will experience minimal weather-related delays. (Schedule)
8. Operational Costs – Estimated construction, maintenance, and staffing costs will remain within the projected budget. (Project Scope & Time-Phased Budget)
9. Launch Demand Growth – Industry demand for additional launch capacity will continue to rise as forecasted. (Project Scope)
10. Infrastructure Scalability – The project design will support future expansion and upgrades without significant redesigns.
11. Regulatory Compliance – All environmental regulations will be met without significant modifications. (Stakeholders List & Schedule)
12. Market Stability – The aerospace market will remain stable, supporting project viability. (Project Scope & Time-Phased Budget)
13. Technological Advancements – Emerging technologies will enhance project efficiency and capabilities. (Resources Chart)
14. Risk Management – Identified risks will be effectively mitigated through proactive measures. (Project Scope & Schedule)
15. Community Support – Local communities will support the project, minimizing opposition and delays. (Stakeholders List)
16. Economic Conditions – Favorable economic conditions will persist, supporting project funding and growth. (Time-Phased Budget)

17. Political Stability – Political conditions will remain stable, ensuring uninterrupted project progress. (Project Stakeholders)
18. Partnership Continuity – Key partnerships will remain intact throughout the project lifecycle. (Project Stakeholders)
19. Innovation Adoption – The Industry will adopt innovative solutions proposed by the project. (Project Scope)
20. Safety Standards – All safety standards will be met, ensuring a secure operational environment. (Resources Chart)
21. Environmental Impact – Environmental impact assessments will show minimal adverse effects. (Schedule)
22. Resource Allocation – Resources will be allocated efficiently to avoid bottlenecks. (Budget)
23. Maintenance Efficiency – Maintenance processes will be streamlined to minimize downtime. (Project Scope)
24. Project Timeline – Project milestones will be achieved within the planned timeline. (Schedule)
25. Cost Control – Effective cost control measures will prevent budget overruns. (Budget)
26. Quality Assurance – Quality assurance processes will ensure high standards are maintained. (Project Scope)
27. Timely Information Sharing:
28. All project-related information will be shared with stakeholders and team members in a timely manner, ensuring everyone is up-to-date with the latest developments. (Stakeholders List & Schedule)

29. Feedback mechanisms will be in place to allow stakeholders and team members to provide input and raise concerns, which will be addressed promptly to maintain project alignment. (Stakeholders List & Human Resources)
30. Communication across all channels will be clear, consistent, and aligned with project goals, minimizing misunderstandings and ensuring everyone is on the same page. (Project Scope & Schedule)

Creative Content

Change Request Form Introduction (Ryan)

The change request form is essential to project management, as it controls the scope and changes. This form ensures that any changes are reviewed and approved before they are implemented on the project.

CHANGE REQUEST FORM

CHANGE DETAILS

Project Name	Aerospace Transport	
Change No.	CR-2025-042	
Change Name	Flame Trench Modification for Launch Pad 2	
Date of Request	3/17/2025	
Requested By	Ryan Plones	
Requester's Contact Information	(419)913-1141	
Date Needed	3/24/2025	

PRIORITY					
x	HIGH		MEDIUM		LOW

Description of Change	Modify the flame trench for launch pad 2 for a more significant thrust from the next-generation rockets, requiring reinforced concrete thickness by 18 inches and expanding the cooling system by 35%
Reason for Change	Research shows that the thrust requirements from the original have increased by 22-30%. Implementing this Change now would avoid costly retrofitting after construction and ensure longevity involving rocket engineering.

CHANGE IMPACTS

Scope	REDESIGN STRUCTURE OF FLAME TRENCH; COOLING SYSTEM REQUIRED
Deliverables	MODIFIED FLAME TRENCH FOR LAUNCH PAD 2
Cost	$876,000 FOR MATERIALS
Resources	80 ADDITIONAL HOURS (ENGINEERING TEAM)
Timeline	14 DAYS
Stakeholders	CONSTRUCTION TEAM, ENGINEERING TEAM, PROJECT SPONSORS

RISK ANALYSIS

Risk Identification	Potential concrete curing complications with increased thickness
Probability of Risk	Medium
Risk Mitigation Strategies	Talk to the concrete guys for the best curing approach and develop a strategy to monitor the plan for thicker sections.

DECISION

x	ACCEPTED
	REJECTED
	More Info Requested

Project Manager Name	Signature	Date
Dezso Kovi	Dezso Kovi	3/20/25

Decision-Maker Name & Title	Signature	Date
Fabian Collin & Engineering Lead	Fabian Collin	3/21/2025

Key takeaways (Ryan)

Change request forms will prevent unnecessary unplanned work from messing up the project flow. Our form will help control technical specifications and budget problems.

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Risk Matrix Introduction (Ryan)

The risk matrix is a tool used to detect or plan for a potential project risk. It does this by measuring the probability and impact. This helps people in the project plan and prepare for a negative consequence.

LIKELIHOOD ↓ / SEVERITY →	1	2	3
1	**LOW** **Material cost estimation**	**LOW** **Workforce availability**	**MEDIUM** **Weather delays**
2	**LOW** **Regulatory approvals**	**MEDIUM** **Material suppliers**	**HIGH** **Engineering design changes**
3	**MEDIUM** **Scope changes**	**HIGH** **Environmental compliance delays**	**HIGH** **Technical integration failures**

Key takeaways (Ryan)

This tool's ability to predict issues early will help the project have less stress and more project protection. This is extremely helpful for a company that deals with things like aerospace.

Request for quote introduction (Ryan)

A request for quote is a document dedicated towards suppliers to negotiate price bids on specific items or a service. This is to help with competitive pricing on things such as large aerospace equipment and the transportation of those goods.

REQUEST FOR QUOTE (RFQ)

[Advanced Aerospace Materials] Date: March 25, 2025

[4200 Propulsion Way Houston, Texas 77058] Quotation: ATP-RFQ-2025-042

[Company Phone: 713-555-8900, Email: sales@advancedaerospace.com] Customer ID: ATP-2025

Quote for: Quote Valid Until: April 25, 2025

[Dezso Kovi] Prepared By: Ujas Patel

[Aerospace Transport]

[500 Launch Complex Drive Cape Canaveral, FL 32920]

[Company Phone: 321-555-7700, Email: procurement@aerospacetransport.com]

Comments or Special Instructions: Request for flame-deflector panels for Launch pads 2. All materials must meet ASTM A240 requirements for 310S stainless steel. Heat resistance certification is required

Salesperson	P.O. Number	Ship Date	Ship Via	F.O.B. Point	Terms
Marcus Johnson	ATP-PO-2025-118	May 15, 2025	FedEx Freight	Cape Canaveral	Net 30

Quantity	Description	Unit Price	Taxable	Amount
24	Flame Deflector Panels, 310S, 1.5" thick, 4'x8', water-cooled	$18,750	Yes	$450,000
96	Mounting Brackets, Grade 8, titanium alloy	$475	Yes	$45,600
			Subtotal	$511,200
			Tax Rate	6.5%
			Sale Tax	$33,228
			Other	$7,500
			Total	$551,928

NOTE: ***All components must be manufactured in accordance with aerospace specifications AS9100D. Material test reports and certification are a must for required deliverables.***

Key takeaways (Ryan)

An official request for proposals (RFQ) makes sure that they choose the right vendors or suppliers of certain goods or services.

Change management communications introduction (Ryan)

For a project as large as ours, we need a clear way to communicate big details and changes. The form will lay out a few things, as you will see here, but most importantly, on the left-hand side, first is the event or action, followed by the phase, and the dates it affects. You can also see the area where it lists target stakeholders and what they are communicating.

CHANGE MANAGEMENT COMMUNICATIONS PLAN

PROJECT NAME	PROJECT MANAGER	VERSION NO.
Aerospace Transport	Fabian Collin	1.0.0

ORGANIZATION	DATE CREATED	VERSION DATE
Aerospace Transport Team	04/27/2025	04/27/2025

Identify **affected stakeholders** and describe required **communication**

EVENT / ACTION / STRATEGY	PROJECT PHASE(S)	EST. DATE OF EFFECT	TARGETED STAKEHOLDERS	REASON FOR COMMUNICATION	METHOD OF COMMUNICATION
Implentation of New Transportation Strategy	2.4 & 2.5	07/17/2025	SpaceX and Blue Origin, Supply chain analysts	Communicate a finalization for transportation appraoch for oversize components and coordinate logistics implentation across teams in all companies involved.	Virtual meeting, Logisitcal schedule dashboard
Performance meteric system launch	5.2	08/27/2025	Project Managers, KPI Dashboard, SpaceX, Blue Origin and NASA	New performance tacking system to ensure that all teams understand how to use and utilize metrics for continous improvement within the project and after project completion.	Training sessions, User manual documentation, Dashboard Demonstation
Launch pad design review workshop	1.3, 2.1	06/15/2025	Engineering teams, Enviromental, Saftety officers and compliance teams	Review all the design specification before manufacturing begins.	In-person design meetings and technical documentation
Supply chain simulation exercise	2.6	07/28/2025	Logistics team, procurement teams, risk management, project manager, supply chain specialist	Contingency plans and response scenarios of supply chain disruptions with a help from supply chain specialist	Simulation, protocol walktrhough, digital scenario testing documentation

Key takeaways (Ryan)

As you can see from the form, it is a structured form of communication, and it is a strong form as well. This also helps us tailor the communication to the right person or group through their preferred form of communication.

Project Closeout Introduction (Ryan)

This form officially marks the closeout of the aerospace project and provides a summary of all the things we have discussed. Some of those things listed are the project's deliverables, expenses, schedule performance, roles and responsibilities, and lessons learned. This form signifies that the project is closing out and that all objectives have been satisfied, final approvals from specified people are documented, and critical insights are captured to improve future project performance on other projects we may pursue.

SIMPLE PROJECT CLOSEOUT REPORT

PROJECT NAME

Aerospace Transport Launchpad Project

PROJECT SPONSOR	PROJECT MANAGER	PROJECT START DATE	PROJECT END DATE
Dr. Eleanor Ramos	Dezso Kovi	January 6th 2025	August 15th 2025

PROJECT SUMMARY

The project successfully delivered multiple launch pads supported by extensive infrastructure and operational systems designed to enhance turnaround time and increase launch frequencies for space exploration. Deliverables included reinforced concrete launch p[ads capable of withstanding rockets up to 2,000 tons, flexible mounting systems adaptable to various rocket configurations, and flame trenches equipped with water cooling systems to manage thermal and acoustic loads during launches. Finally, the project provided a logistical infrastructure to secure rocket fuel storage and transportation and integrated systems to enable real-time tracking, monitoring, and management of launch activities. The outcomes were developed and aligned with our project objectives to help improve operational efficiency and support higher-volume launches.

PROJECT ROLES & RESPONSIBILITIES

NAME	ROLE	RESPONSIBILITIES
Dezso Kovi	Project Manager	The overall project leadership, scope, stakeholder coordination, final reporting, and budget oversite
Ryan Plones	Schedule Manager	Task duration estimation, critical path identification, schedule, and monitoring project timeline.
Christian Jackson	Project Schedule Coordinator	Gantt chart manager, budget, timeline coordination, and assumptions management
Ujas Patel	Financial Analyst	Time-phased budget creation, budget line chart analysis, stakeholder analyzation, and developing contingency plans
Fabian Collin	Supply Chain Specialist	Production planning, transportation logistics and identifying manufacturers, and assumption management
Nathen Flies	Communications Manager	Communication needs assessment, stakeholder communications planning, regular status updates, and documentation management.
Dr. Eleanor Ramos	NASA Space Launch Director	Executive oversight, regulatory coordination, government agency liaison, and final approval authority
Thomas Clarke	SpaceX VP of Launch Operations (Executive Stakeholder)	Commercial spaceflight expertise, operational standards consultation, and industry best practices guidance

DELIVERABLES

PLANNED	ACTUAL	COMMENTS
Laund pad reinforced concrete up to 2,000 tons, flexible mountain points, and flame trenches.	Completed with a 99.8% compliance specification	Reinforced concrete will help meet the load requirement by 5%. Have a water-cooled system to help perform better testing.
Site development	Completed with minor modification	Environmental protection measures in place during construction to address any stakeholder concerns as well as community
Logistics infrastructure	Completed with all the key systems and fully operational	The inventory tracking system has exceeded expectations and helped reduce overhead by 35%
Operational Systems	Systems deployed and tested	AI-enhanced analytics tools to help with additional capabilities

EXPENSES

PROJECT PHASE	PLANNED BUDGET	ACTUAL COSTS	COMMENTS
Production & Transportation (2.1-2.7)	$48,000.00	$54,847.86	14.3% over budget due to higher material costs for simulation software and route optimization tools.
Trail Runs (5.1-5.3)	$29,000.00	$33,436.24	15.3% over budget due to the business tools and survey tools needed for more heavy comprehensive testing
Weekly Updates (6.1)	$1,200.00	$1,446.00	There is a 20.5% increase because of KPI dashboard development and reporting infrastructure.
Project total	$109,000.00	$137,697.00	26.3% over budget because necessary material costs were underestimated in the planning phase.

SCHEDULE

KEY MILESTONES	INITIAL DEADLINE	ACTUAL COMPLETION	COMMENTS
Initial Assessment & Partner Engagement	March 15th	March 20th	Delay due to additional stakeholder consultation but provided valuable insight.
Production Planning & Supply Chain Optimization	April 30th	April 28th	Ahead of schedule through parallel processing of manufacturing requirements.
Transportation Strategy Implementation	June 15th	June 18th	Weather delays affected route testing, but a contingency plan was activated successfully.
Final testing & Project Handover	August 15th	August 15th	The deadline was met after the team implemented final testing.

LESSONS LEARNED / RECOMMENDATIONS

Our project exposed significant gaps in our standard approach to infrastructure development. Material costs exceeded plans by 26.3%, primarily due to insufficient contingency buffers for specialized components. Experienced production delays. The response team we assembled in March demonstrated problem-solving capacity, resolving technical integration challenges that traditional approaches would have extended by weeks. We implemented KPI dashboards too late in the cycle, missing early warning indicators for the testing phase bottlenecks. Weather contingencies consumed 40% of our buffer time, suggesting our standard 15% allocation is insufficient for exposed construction elements. The 14% over budget of the documentation system investment reduced validation time by 32% and should become standard practice.

APPROVAL

NAME	TITLE	SIGNATURE	DATE
Dezso Kovi	Project Manager	Dezso Kovi	4/29/25

NAME	TITLE	SIGNATURE	DATE
Dr. Elenor Ramos	NASA Space Launch Director	Elenor Ramos	4/29/2025

NAME	TITLE	SIGNATURE	DATE
Thomas Clarke	SpaceX VP of Launch Operations	Thomas Clarke	4/29/2025

Key takeaways (Ryan)

This form will help us show that we had clear planning and used quick problem-solving to help deliver this project effectively. It will also help us reflect on what we might have done wrong in our first go-rounds.

Appendices

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Team Contribution

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Section Number	**#0003**
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Ryan Plones	Schedule management plan, Scope Management Plan, Creative content, Grammar Checking
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